Question 1

Planning (15 marks)

Defining the problem (3 marks)

P1 *V*ary *V* or *f* [1]

P2 Measure *f* for different *V* or measure *V* for different *f* [1]

P3 Keep temperature constant [1]

Methods of data collection (5 marks)

M1 labelled diagram including source of sound adjacent to the opening e.g. loudspeaker/tuning fork [1]

M2 Method of producing sound of different frequencies e.g. several tuning forks or signal generator [1]

M3 Method of measuring volume of air – volume of container - volume of water or find total volume of each different container [1]

M4 Method of determining resonant frequency e.g. largest sound heard or displayed [1]

M5 Perform experiment in quiet room or avoid other noise [1]

Method of analysis (2 marks)

A1 Plot a graph of f^2 against $1/V$ or lg f against lg V or or lg f against lg $1/V$ [1]

A2 Relationship is correct if graph is a straight line through the origin or straight line for log-log graph [1]

Safety considerations (1 mark)

S Switch off power supply when not in use/ ear defenders for loudspeaker method [1]

Additional detail (4 marks)

D Relevant points might include [4]

1. Detail on measuring volume – use of measuring cylinder/burette
2. Determination of frequency using oscilloscope/read off tuning fork or signal generator
3. Detailed timebase calculation
4. Detail determining resonance e.g. adding/subtracting small amounts of water/changing signal generator to create resonance
5. Discussion of container e.g. end correction/shape of mouth of bottle
6. Gradient = k or lg f = -0.5 lg V + 0.5 lg k or lg f = 0.5 lg $1/V$ + 0.5 lg k
7. Constant amplitude/intensity of source of sound
8. Method to check fundamental frequency.

15 marks can be scored in total.

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